

REVU KAVITHA

[revukavitha](#) | [in revukavitha](#) | [myportfolio](#) | [revukavitha789@gmail.com](#)

SUMMARY

Software Engineer with strong foundations in Data Structures, Computer Networks, and backend development. Experienced in building scalable systems, real-time monitoring applications, and AI-powered solutions. Passionate about designing efficient, network-aware systems with a focus on performance, reliability, and real-world impact.

WORK EXPERIENCE

Web Developer Intern

HCL Technologies

- Developed responsive frontend components and integrated backend APIs for dynamic data handling.
- Gained experience in building scalable web applications and working with client-server architecture in a structured development environment.

Student Volunteer - Debugging & Mobile App Contest

IEI Puducherry PTU

- Assisted in organizing a technical contest focused on debugging and problem-solving.
- Helped participants resolve coding issues and implementation challenges.
- Collaborated with teams to ensure smooth execution of the event.

PROJECTS

NetPulse: Network Monitoring & API Performance Analyzer

[Live Demo](#) | [GitHub](#)

- Designed a network monitoring system to analyze API and website performance by tracking response latency, status codes, and uptime in real time.
- Implemented periodic health checks and visualized system behavior through an interactive dashboard.
- Concepts: HTTP Lifecycle, Latency Analysis, API Monitoring, Client-Server Architecture | Tech: FastAPI, Next.js.

NetScope: HTTP Traffic Analyzer & Monitoring Dashboard

[Live Demo](#) | [GitHub](#)

- Built a real-time HTTP traffic monitoring system to capture and analyze request flows, compute latency, and track system performance.
- Implemented middleware-based logging and live dashboard visualization for request insights.
- Concepts: Request-Response Lifecycle, Middleware, System Monitoring | Tech: FastAPI, Next.js.

RAG Knowledge Assistant

[Live Demo](#) | [GitHub](#)

- Designed a document-based question-answering system to retrieve and generate context-aware responses from large unstructured datasets.
- Implemented semantic search using vector embeddings to improve retrieval accuracy and reduce irrelevant responses.
- Built scalable APIs to handle real-time queries with optimized response latency. Concepts: Information Retrieval, Embeddings, API Design, Latency Optimization | Tech: FastAPI, LangChain, Vector Database.

Multi-Model AI Orchestrator

[Live Demo](#) | [GitHub](#)

- Developed a system to evaluate and dynamically route requests across multiple AI models based on latency, cost, and response quality.
- Designed benchmarking metrics and scoring mechanisms to optimize model selection and improve response efficiency in real-time systems.
- Concepts: System Optimization, Model Routing, Performance Evaluation | Tech: Python, APIs.

Deep Learning Model Lab

[Live Demo](#) | [GitHub](#)

- Built a multimodal AI system for image-based question answering, focusing on efficient inference and model performance.
- Optimized processing pipelines to reduce latency and improve response time for real-world usage scenarios.
- Concepts: Deep Learning, Inference Optimization, Multimodal Systems | Tech: Hugging Face Transformers, Python.

EDUCATION

- M.Tech**, Puducherry Technological University - CGPA: 9.58 (Ongoing)
- B.Tech**, Puducherry Technological University - CGPA: 8.13
- 12th Class**, Sri Sai Junior College, Yanam - CGPA: 8.0
- 10th Class**, Dr. K. R. Narayanan Govt High School - GPA: 9.2

CERTIFICATIONS

- Udemy Introduction to Computer Networks Certification in networking fundamentals.
- Udemy Hadoop Certification in distributed data processing.
- Udemy SQL Bootcamp Certification in database querying and optimization.
- Udemy AI Engineer Production Track in LLM and agent deployment.
- Udemy Linux Certification in practical system administration.
- Diploma in Computer Applications, Technical Program.

ACHIEVEMENTS

- Cleared the national cutoff in Joint Entrance Examination (JEE) Mains 2020.
- Selected to represent at the Regional Level Science Exhibition (2017-18), Government of Puducherry.

SKILLS

- Programming:** Python, JavaScript
- Core CS:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks (TCP/IP, HTTP, DNS)
- Backend & Systems:** FastAPI, REST APIs, System Design Basics
- Databases:** MongoDB, SQL, Neo4j (Graph DB), Cassandra (NoSQL Columnar)
- Networking Concepts:** HTTP Lifecycle, Latency Analysis, API Monitoring, Client-Server Architecture
- AI/ML:** Scikit-learn, NLP, LLMs, RAG
- Tools:** Git, Docker